STEP 1 — Create Notebook

```
#Lab10.2_LogisticRegression_v.sravya_2403A52403.ipynb
```

STEP 2 — Import Libraries

```
# Data handling
import numpy as np
import pandas as pd

# Text preprocessing
import re
import nltk
from nltk.corpus import stopwords

# Feature extraction
from sklearn.feature_extraction.text import TfidfVectorizer

# Model training
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split

# Evaluation
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Visualization
import matplotlib.pyplot as plt
import seaborn as sns
```

STEP 3 — Load and Preprocess Data

```
# Example loading
data = pd.read_csv("/content/IMDB Dataset.csv", engine='python', on_bad_lines='warn')

# Download stopwords (first time only)
nltk.download('stopwords')
stop_words = set(stopwords.words('english'))

def clean_text(text):
    text = text.lower()
    text = re.sub(r'[^a-zA-Z]', ' ', text)
    words = text.split()
    words = [word for word in words if word not in stop_words]
    return " ".join(words)

data['clean_review'] = data['review'].apply(clean_text)

print(data[['review','clean_review']].head())
```

```
/tmp/ipython-input-2476085702.py:2: ParserWarning: Skipping line 4786: unexpected end of data

  data = pd.read_csv("/content/IMDB Dataset.csv", engine='python', on_bad_lines='warn')
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
                                              review  \
0  One of the other reviewers has mentioned that ...
1  A wonderful little production. <br /><br />The...
2  I thought this was a wonderful way to spend ti...
3  Basically there's a family where a little boy ...
4  Petter Mattei's "Love in the Time of Money" is...

                                        clean_review
0  one reviewers mentioned watching oz episode ho...
1  wonderful little production br br filming tech...
2  thought wonderful way spend time hot summer we...
3  basically family little boy jake thinks zombie...
4  petter mattei love time money visually stunnin...
```

STEP 4 — Feature Extraction (TF-IDF)

```
tfidf = TfidfVectorizer(max_features=5000)

X = tfidf.fit_transform(data['clean_review'])
y = data['sentiment'] # Changed 'label' to 'sentiment'

print("Vocabulary Size:", len(tfidf.vocabulary_))
```

```
Vocabulary Size: 5000
```

STEP 5 — Train Logistic Regression Model

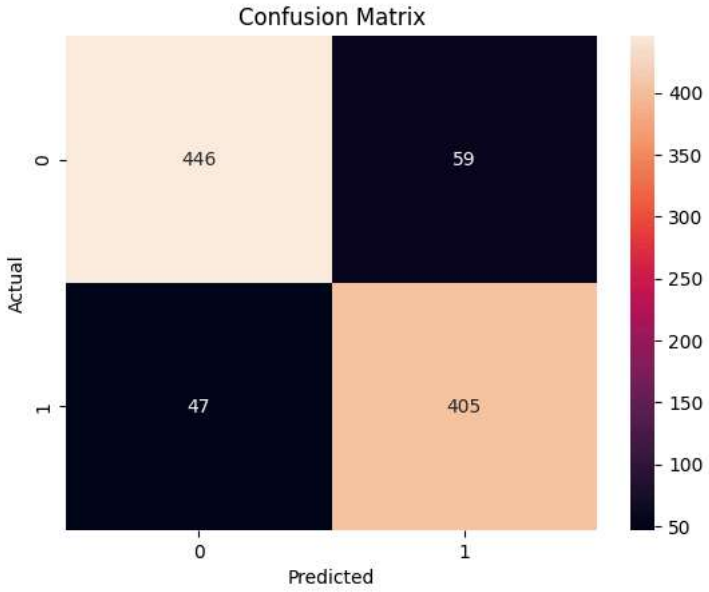
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)
```

```
▾ LogisticRegression
LogisticRegression(max_iter=200)
```

STEP 6 — Model Evaluation

```
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))

cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```

```
Accuracy: 0.8892371995820272
              precision    recall  f1-score   support

    negative       0.90      0.88      0.89       505
    positive       0.87      0.90      0.88       452

    accuracy                           0.89       957
   macro avg       0.89      0.89      0.89       957
weighted avg       0.89      0.89      0.89       957
```



STEP 7 — Analysis (Comparison with Naive Bayes) Here is a sample 8–10 sentence analysis:

```
# Logistic Regression is a discriminative model, while Naive Bayes is a generative model.

# Logistic Regression directly models P(y|x)P(y|x)P(y|x), whereas Naive Bayes models P(x|y)P(x|y)P(x|y). # Fixed: Changed '|' to

# Logistic Regression usually performs better when features are correlated.

# Naive Bayes assumes feature independence, which is unrealistic in text data.

# Logistic Regression gives better decision boundaries.

# Naive Bayes trains faster on large datasets.

# Logistic Regression may require more iterations to converge.

# Logistic Regression often achieves higher accuracy with TF-IDF features.

# Naive Bayes works well as a baseline model.

# In sentiment analysis, Logistic Regression typically provides better precision and F1-score.
```